

Bullet Hole Detection

~~Current state of the vision model~~

What it detects reliably today, where it fails,
and what is not yet measured.

What the system reports

- It finds the paper Targets on a Board in range camera footage, then the bullet holes around them, and reports each hole's distance from the Target centre.
- It counts visible marks on paper — not rounds fired. Two rounds through nearly the same point leave one mark and are reported as one.
- That is a limit of single-frame vision, not a defect: the information needed to separate them is not in the picture. Telling them apart would need shot-by-shot video tracking, which the system does not do today.

Read a reported count as “holes visible on the paper”, never as “rounds fired”.

Detection settings are now tuned per camera setup

Hole size in the picture depends on zoom, distance and calibre, so one fixed setting cannot serve every lane. Settings are now saved per setup, and duplicate detections are merged by a single rule instead of four conflicting ones. Same board, same model, same photograph:

Previous fixed setting — 2 holes reported



Tuned for this setup — 17 holes reported



A person counts roughly 16 holes on this board.

WEAKNESS 1

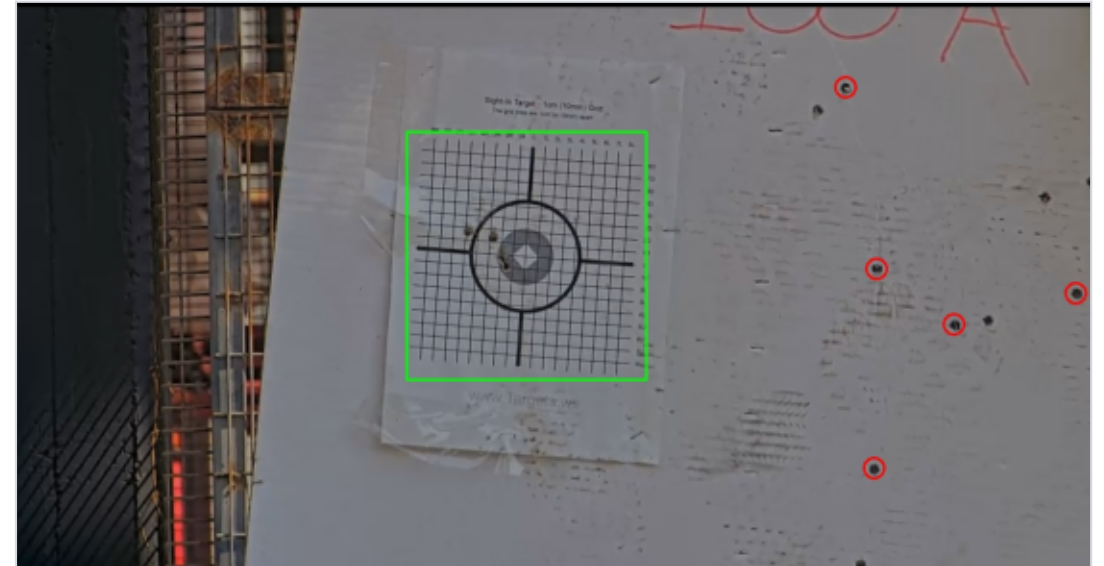
Each new camera setup needs tuning, and close-ups stay weak

Tuning is manual, one-off work per setup, and it does not rescue every capture. This close-up frame reports nothing at the standard setting. Retuned, it finds 5 — while roughly 15 marks are visible on the board, and the holes inside the Target itself are still missed.

Standard setting — 0 holes reported



Retuned for this setup — 5 holes reported



Across 13 sample captures, 5 returned zero holes at the shipped default setting.

WEAKNESS 2

Boards far from the camera are not detected at all

The system has to find the Target before it can count holes around it. On distant boards the Target is never found, so the board reports nothing — even where a viewer can plainly see the holes.

Only boards near the camera are reliable today.

Two distant boards — no Target found on either, so no holes are reported



What this means for a session

- A lane whose boards sit beyond the reliable range yields empty results rather than a warning that nothing was found.
- Camera placement therefore decides whether a session produces usable data at all.

WEAKNESS 3

Strong on its own data, well short of it in the field

The model scores highly when graded against the dataset it was trained on. On real range captures it does not hold up. That gap — not the scoring — is the honest picture of where it stands.

Graded on its own dataset

held-out split of the training footage, Aug 2025

F1	0.90
Precision	0.90
Recall	0.90
mAP@50	0.93

On real range captures

no equivalent score exists — nothing is labelled

- Reference board: 2 holes found of ~16 visible, at the shipped setting
- Close-up capture: 5 found of ~15 visible, even after retuning
- 5 of 13 sample captures returned nothing at all

The two columns are not like for like: the right-hand figures are eyeball counts, not a measured score.

WHAT IS MISSING

Field accuracy has never been measured

- The 0.90 above was earned on footage from the same source as the training images. It says the model learned that material, not how it does on a lane it has not seen.
- Nothing on the range side is labelled, so two things stay unknown: how many real holes are missed, and how many reported holes are not holes at all.
- It also means nothing would detect a silent drop in quality after a settings change or a model update.

To close it

Hand-label the holes on a set of captures representing the real lanes, distances and calibres. That fills the right-hand column with a real number, turns tuning into evidence rather than judgement, and provides a regression check that runs on every change.